Four-Stream MSGL-Transformer for Mouse Social Behavior Recognition

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Problem Statement

◆Manual annotation of rodent social behavior is slow, subjective, and does not scale to modern experiments ◆Existing methods require explicit skeleton estimation or fuse heterogeneous feature streams independently ◆Prior methods on the CRIM13 dataset use non-standard evaluation splits, making fair comparison difficult

Our goal is to build a multi-stream transformer that jointly fuses trajectory and spatio-temporal features from two camera views *without* skeleton estimation

Contributions

◆**Extended** the single-stream MSGL-Transformer to a **four-stream architecture** processing trajectory-short, trajectory-long, TOP-view, and SIDE-view MARS features ◆**Applied cross-stream attention** to capture complementary interactions between trajectory pairs and synchronized camera view pairs ◆**Employed** an **adaptive sigmoid fusion gate** that dynamically learns per-stream contribution weights ◆**Achieved 68.83% avg per-class accuracy** and **71.0% overall accuracy** on the official CRIM13 split (+7.63 pp)

CRIM13 Dataset

- **237 videos** of resident-intruder mouse pairs from top and side cameras at 25 fps
- **Official split:** 104 training / 133 test videos
- **12 behaviors:** approach, attack, mount, chase, circle, drink, eat, clean, sniff, up, walk, other
- Severe imbalance: *other* = 58.2%
- **32,201** train / **39,338** test sequences

Input Representation

- **Trajectory-short** ($\mathbb{R}^{100\times2851}$): 75-frame (3 s) window
- **Trajectory-long** ($\mathbb{R}^{100\times2851}$): 615-frame (25 s) window
- **TOP-view ST** ($\mathbb{R}^{100\times980}$): top camera descriptors
- **SIDE-view ST** ($\mathbb{R}^{100\times980}$): side camera descriptors

No skeleton estimation required; pre-extracted by MARS.

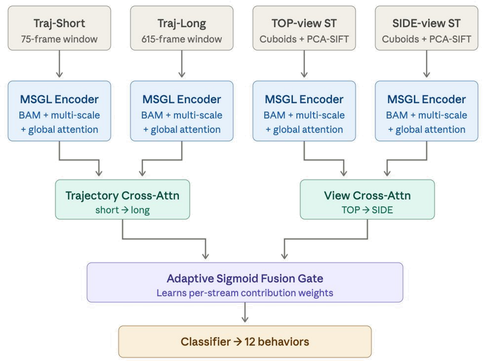
Architecture Components

- **BAM:** sigmoid-gated channel reweighting for discriminative temporal regions
- **Multi-scale local attention:** short (50 fr) and medium (100 fr) causal
- **Global self-attention:** full-sequence including CLS token
- **Cross-stream attention:** trajectory short→long; TOP→SIDE view
- **Adaptive fusion gate:** sigmoid MLP learns per-stream weights

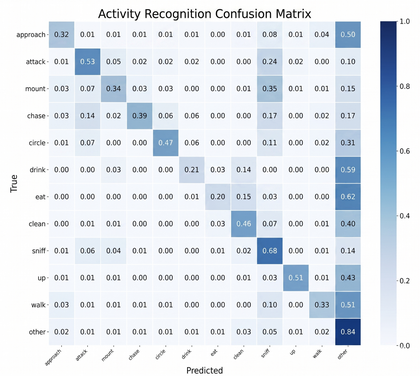
Training Configuration

- **Loss:** weighted cross-entropy (inverse-frequency)
- **Optimizer:** AdamW, lr = 10^{-4}, cosine annealing
- **Batch:** 32 **Dropout:** 0.2 **Params:** 2.9M
- **Early stopping:** patience 15; best at epoch 27

Proposed Architecture



Confusion Matrix



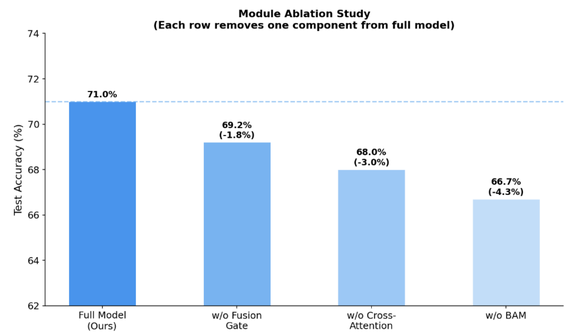
- Main confusions: *sniff*↔*other*, *eat*↔*clean*
- *Eat* confused with *clean* due to similar stationary motion patterns;
- *Eat*, *drink*, *circle* most challenging as indicated by low diagonal values

Main Results

Method	Avg/Cls	Overall
Human expert	—	~70.0%
Burgos-Artizzu et al. (2012)	61.2%	—
LSTM	—	62.3%
GRU	—	63.5%
BiLSTM	—	63.9%
TCN	—	68.2%
Transformer	—	68.9%
Ours (MSGL-Transformer)	**68.83%**	**71.0%**

- **+7.63 pp** over handcrafted baseline; **+2.1 pp** over plain Transformer
- Approaches human inter-annotator agreement (~70%)
- Inference: **0.59 ms/sequence** on RTX A4000

Ablation Studies



- BAM: +4.3%, cross-attention: +3.0%, fusion gate: +1.8%
- Trajectory-short most important (−5.3% when removed)
- Single-stream TOP-only: 54.3% vs. 71.0% full model

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